Fairfax Invitational - Repaired practice edition - Answer key

Division C | Forensics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

chemistry-7. (1 points) Authored replacement for source item chemistry-7: Repair data: a sodium carbonate solution has pH 11.2; a calcium nitrate solution has pH 6.8. Which of these two solutions is basic, and what threshold did you use?

answer: Sodium carbonate; pH 11.2 is above 7 under the stated ordinary aqueous comparison. The supplied nitrate solution is below 7. Only these two measured solutions are being compared. (1 points).

mass-1. (1 points) Authored replacement for source item mass-1: The mass spectrum has its tallest peak at m/z 31 and a smaller molecular-ion peak at m/z 46. What is the m/z of the base peak?

answer: 31 (1 points).

Numerical target 31; absolute tolerance 0.062. Accepted units: , .

mass-2. (1 points) Authored replacement for source item mass-2: For a singly charged molecular ion with m/z 46 and a fragment base peak at m/z 31, what is the nominal molecular mass?

answer: 46 Da (1 points).

Numerical target 46; absolute tolerance 0.092. Accepted units: , Da.

repair-case. (4 points) Authored written substitute for unavailable source material: Why might a well-supported multilocus DNA comparison be more discriminating than a common fiber-class agreement?

answer: It can provide more specific source information when evaluated with valid statistics and quality assumptions; relevance and transfer still matter. (4 points).